

CyberSecurity Lab1

Part 1 : understanding Password Security

Step1 : Analyse Weak Passwords

Discussion

1. Why are these passwords insecure?

Passwords such as 123456, password, admin, 12345678, and qwerty are insecure because they are very common and easy to guess. They are short and do not contain a strong mix of uppercase letters, lowercase letters, numbers, and special characters. Because many people use them, attackers can easily guess them using automated tools.

2. How quickly could a computer guess these passwords?

A computer can guess these passwords very quickly, sometimes in a few seconds. Attackers use automated tools that test thousands or even millions of password combinations in a short time. Since these passwords are common, they are usually cracked almost immediately.

3. Attackers often use automated tools to try many passwords quickly. A dictionary attack tests a list of common passwords, while a brute force attack tries many possible combinations until the correct password is found. These methods make weak passwords easy to crack.

Step 2: Create Strong Passwords

Let's write my examples below:

1. Cyb3r\$Shield_92
2. Str0ng!P@ssKey45
3. Gi\$ecur3_Lab88

Part 2: Observing Network Traffic

The Wireshark Network Analyzer

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

Bienvenue dans Wireshark

Capture

...en utilisant ce filtre : Toutes les interfaces affichées

Wi-Fi

☒ Connexion au réseau local* 10

☐ Connexion au réseau local* 9

☐ Connexion au réseau local* 8

☐ Cellulaire

☐ Connexion réseau Bluetooth

☐ Connexion au réseau local* 2

☐ Connexion au réseau local* 1

☐ Adapter for loopback traffic capture

☐ Ethernet

☒ Event Tracing for Windows (ETW) reader: etwdump

Découvrir

Guide Utilisateur · Wiki · Questions et Réponses · Listes de diffusion · SharkFest · Discord Wireshark · Faire un don

Vous exécutez Wireshark 4.6.4 (v4.6.4-0-g93282876538d). Vous recevez les mises à jour automatiques.

Wireshark Network Analyzer

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Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
31623	51.708269	172.64.155.209	192.168.100.121	UDP	66	443 → 60345 Len=24
31624	51.708269	172.64.155.209	192.168.100.121	UDP	66	443 → 60345 Len=24
31625	51.708269	2.19.251.114	192.168.100.121	TCP	54	443 → 5634 [RST] Seq=5792 Win=0 Len=0
31626	51.708269	2.19.251.114	192.168.100.121	TCP	54	443 → 5633 [RST] Seq=5791 Win=0 Len=0
31627	51.708269	192.168.100.121	192.168.100.1	TCP	66	[TCP Dup ACK 31619] Seq=452 [ACK] Seq=6225 Ack=31377 Win=63792 Len=0 Sli=15488 SRO=16861
31628	51.708269	192.168.100.121	192.168.100.1	TCP	66	[TCP Dup ACK 31619] Seq=452 [ACK] Seq=6225 Ack=31377 Win=63792 Len=0 Sli=16861 SRO=16873
31629	51.754685	192.168.100.121	192.168.100.1	DNS	72	Standard query 0x3a7f HTTPS www.bing.com
31630	51.755481	192.168.100.121	192.168.100.1	DNS	72	Standard query response 0x3a7f HTTPS www.bing.com
31631	51.800812	192.168.100.121	192.168.100.121	TCP	1466	443 → 5638 [ACK] Seq=679627 Ack=5761 Win=48848 Len=1412
31632	51.834088	192.168.100.121	192.168.100.1	UDP	83	59838 → 443 Len=41
31633	51.849746	192.168.100.121	192.168.100.1	TCP	54	5626 → 443 [ACK] Seq=6761 Ack=55829 Win=587896 Len=0
31634	51.850808	192.168.100.121	192.168.100.121	QUIC	1292	Initial, DCID=11ce0b771fcc7f, PDR: 6, PADDING, PING, PING, PADDING, PING, PADDING
31635	51.977893	192.168.100.121	172.64.155.209	UDP	1292	68345 → 443 Len=1250

Frame 30160: Packet, 1446 bytes on wire (11568 bits), 1446 bytes captured (11568 bits) on interface 0

Ethernet II, Src: HuaweiTechno_38:6a:5f (48:ac:29:38:6a:5f), Dst: Intel_55:c1:21 (f8:00:27:55:c1:21)

Internet Protocol Version 4, Src: 172.64.155.209, Dst: 192.168.100.121

Transmission Control Protocol, Src Port: 443, Dst Port: 5623, Seq: 3642886, Ack: 3686,

0000 40 05 0f a5 c5 21 48 4c 29 38 6a 5f 08 00 45 00 ... IHL 38|...E|

0010 05 08 3a 07 48 00 4f 86 4c 07 38 00 27 20 c0 a8 ... 4 0 0 1 8 7 ...

0020 64 79 01 3b 15 f7 97 82 39 0f b3 76 2f 07 50 18 ... 0 y ... 9 v / p ...

0030 01 24 05 05 00 00 f7 50 2c c1 02 8a c1 37 51 a0 ... 5 ... 4 8 ... 7 ...

0040 f2 be 72 7a 3c fe 06 39 07 c1 03 71 c4 2c 64 fa ... r t x ... 9 ... q ... d ...

0050 71 10 15 a0 50 20 63 91 f7 5c a9 b1 69 0f 8c 08 ... 6 ... P a c ... \ ... 1 ...

0060 11 13 4d ec 07 38 79 45 80 63 aa e8 3d c8 40 ca ... 0 ... R p t ... c ... = ...

0070 a1 af c3 ef 35 b3 5d c8 67 a4 5a 50 c8 43 3c 35 ... 5 ... 5 ... g ... T ... C ... 5 ...

0080 19 af 14 15 42 8a d8 0e ee 02 25 18 f0 55 65 90 ... 50 ... 10 0e ...

0090 05 08 a5 34 7a fa 5b 2e 21 73 e7 c8 a2 ae 19 f5 ... 4 ... [... 1 s ...

0100 16 af fa 4b 36 f8 ba ac 23 65 91 0f 81 c7 12 08 ... 7a ... 0e ...

0110 d1 ff 38 78 de 36 c7 86 3d 6a d2 c3 05 06 4b c8 ... 8 ... 6 ... a ... 3 ...

0120 61 be e7 22 a5 8a a4 74 e7 c8 00 3e 27 be 5e 28 ... a ... 1 ... 5 ... 9 7 ... 4 ...

0130 c2 ef fe 74 44 d0 8d 82 08 f8 20 84 2a 28 3a 31 ... [... 1 ... 5 ... 9 7 ... 4 ...

0140 00 a0 4c 07 28 fa 5a 05 05 8d 88 5f 2d ca f2 5b ... [... 1 ... 2 ... 7 ... 0 ...

0150 1b 4c fe 78 91 53 4e 8a c1 f2 5a de 59 6d 08 97 ... 1 p 204 ... 2 70e ...

0160 3f 72 74 4f f8 70 09 6e 6d ba 70 89 05 3f 14 06 ... 7 r 0 d p ... 7 ... m ... p ... 7 ...

0170 91 c8 05 32 03 fa 43 88 ca a5 51 a0 7f 7a c7 87 ... 2 ... 2 ... 0 ... 0 ...

0180 4a 7a 37 4f af 7a a8 2f 14 63 07 af a0 21 1c 18 ... 1 ... 1 ... 0 ... 0 ...

Part 3: Packet Analysis

Step 1: DNS Traffic

Let's write one domain name observed:

www.google.com

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dns

No.	Time	Source	Destination	Protocol	Length	Info
9670	118.012524	192.168.100.121	192.168.100.1	DNS	79	Standard query 0xe474 A accounts.google.com
9673	118.030986	192.168.100.1	192.168.100.121	DNS	90	Standard query response 0x741e A www.google.com A 172.217.171.36
9674	118.030986	192.168.100.1	192.168.100.121	DNS	343	Standard query response 0x1dbb A accounts.google.com A 173.194.76.84 NS ns4.google.com NS ...
9675	118.030986	192.168.100.1	192.168.100.121	DNS	129	Standard query response 0xafec HTTPS accounts.google.com SOA ns1.google.com
9676	118.054612	192.168.100.1	192.168.100.121	DNS	95	Standard query response 0xe474 A accounts.google.com A 173.194.76.84
9824	119.103322	192.168.100.121	192.168.100.1	DNS	74	Standard query 0x2c9c A www.google.com
9825	119.103388	192.168.100.121	192.168.100.1	DNS	74	Standard query 0x57c2 A ogs.google.com
9826	119.106837	192.168.100.121	192.168.100.1	DNS	74	Standard query 0xdf9e HTTPS ogs.google.com
9827	119.107522	192.168.100.121	192.168.100.1	DNS	74	Standard query 0xadf3 A ogs.google.com
9828	119.213069	192.168.100.1	192.168.100.121	DNS	90	Standard query response 0x2c9c A www.google.com A 172.217.171.36
9829	119.213069	192.168.100.1	192.168.100.121	DNS	359	Standard query response 0x57c2 A ogs.google.com CNAME www3.l.google.com A 142.251.142.142 ...
9830	119.213069	192.168.100.1	192.168.100.121	DNS	359	Standard query response 0xadf3 A ogs.google.com CNAME www3.l.google.com A 142.251.142.142 ...
9831	119.213069	192.168.100.1	192.168.100.121	DNS	145	Standard query response 0xdf9e HTTPS ogs.google.com CNAME www3.l.google.com SOA ns1.google...
9862	119.725829	192.168.100.121	192.168.100.1	DNS	75	Standard query 0x086f HTTPS ssl.gstatic.com
9863	119.726813	192.168.100.121	192.168.100.1	DNS	75	Standard query 0x9dc2 A ssl.gstatic.com

Step 2: HTTP Traffic

The image shows a Wireshark packet capture analysis of an HTTP GET request. The packet list on the left shows a GET request to /gsr1/gsr1.cr1 HTTP/1.1. The packet details pane on the right shows the structure of the HTTP request, including the status bar at the bottom indicating 200 OK (text/html).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.100.121	142.250.74.67	HTTP	261	GET /gsr1/gsr1.cr1 HTTP/1.1
2	0.001818	142.250.74.67	192.168.100.121	HTTP	280	HTTP/1.1 304 Not Modified
3	0.003636	192.168.100.121	142.250.74.67	HTTP	256	GET /r/gsr1.cr1 HTTP/1.1
4	0.005454	142.250.74.67	192.168.100.121	HTTP	281	HTTP/1.1 304 Not Modified
5	0.007272	192.168.100.121	142.250.74.67	HTTP	254	GET /r/r4.cr1 HTTP/1.1
6	0.009090	142.250.74.67	192.168.100.121	HTTP	280	HTTP/1.1 304 Not Modified
7	0.010909	192.168.100.121	142.250.74.67	HTTP	254	GET /r/r1.cr1 HTTP/1.1
8	0.012727	142.250.74.67	192.168.100.121	HTTP	280	HTTP/1.1 304 Not Modified
9	0.014545	192.168.100.121	184.28.84.204	HTTP	281	GET / HTTP/1.1
10	0.016363	184.28.84.204	192.168.100.121	HTTP	317	HTTP/1.1 304 Not Modified
11	0.018181	192.168.100.121	23.207.17.2	HTTP	317	GET /DigiCertHighAssuranceEVRootCA.cr1 HTTP/1.1
12	0.020000	23.207.17.2	192.168.100.121	HTTP	285	HTTP/1.1 304 Not Modified
13	0.021818	192.168.100.121	23.207.17.2	HTTP	307	GET /DigiCertGlobalRootG2.cr1 HTTP/1.1
14	0.023636	23.207.17.2	192.168.100.121	HTTP	284	HTTP/1.1 304 Not Modified
15	0.025454	192.168.100.121	23.207.17.2	HTTP	307	GET /DigiCertGlobalRootG3.cr1 HTTP/1.1
16	0.027272	23.207.17.2	192.168.100.121	HTTP	285	HTTP/1.1 304 Not Modified
17	0.029090	192.168.100.121	2.19.251.139	HTTP	136	GET /ncc.txt HTTP/1.1
18	0.030909	2.19.251.139	192.168.100.121	HTTP	205	HTTP/1.1 200 OK (text/html)

Packet 18 details:

- Frame 1851: Packet, 205 bytes on wire (1640 bits), 205 bytes captured (1640 bits) on 1
- Ethernet II, Src: HuaweiTechno_38:6a:5f (48:4c:29:38:6a:5f), Dst: Intel_a5:c5:21 (f0:c0:64:79:00:04)
- Internet Protocol Version 4, Src: 2.19.251.139, Dst: 192.168.100.121
- Transmission Control Protocol, Src Port: 80, Dst Port: 1261, Seq: 1, Ack: 83, Len: 151
- Hypertext Transfer Protocol
- Line-based text data: text/html (1 lines)

Packet 18 status bar: 200 OK (text/html)

Step 3: Encrypted Traffic

The figure displays a Wireshark network traffic analysis window. The top pane shows a list of captured packets, highlighting several TLSv1.3 packets related to a handshake. The bottom pane provides a detailed view of the selected packet (No. 5798), showing the raw bytes and their hexadecimal representation.

No.	Time	Source	Destination	Protocol	Length	Info
5724	85.729033	192.168.100.121	192.168.100.121	TLSv1.3	12762	Application Data, Application Data, Application Data, Application Data, Application Data, ...
5725	85.729921	192.168.100.121	192.168.100.121	TLSv1.3	11350	Application Data, Application Data, Application Data, Application Data, Application Data
5737	85.755197	192.168.100.121	192.168.100.121	TLSv1.3	4290	Application Data
5739	85.756706	192.168.100.121	192.168.100.121	TLSv1.3	16998	Application Data, Application Data, Application Data, Application Data
5740	85.756706	192.168.100.121	192.168.100.121	TLSv1.3	1434	Application Data, Application Data
5748	85.836582	192.168.100.121	192.168.100.121	TLSv1.3	161	Application Data
5752	85.945539	192.168.100.121	192.168.100.121	TLSv1.3	395	Application Data
5753	85.945539	192.168.100.121	192.168.100.121	TLSv1.3	513	Application Data
5754	85.945539	192.168.100.121	192.168.100.121	TLSv1.3	85	Application Data
5768	86.100804	192.168.100.121	192.168.100.121	TLSv1.3	373	Client Hello (SHAJm-n-a.googleapis.com)
5779	86.304718	192.168.100.121	192.168.100.121	TLSv1.3	1470	Server Hello, Change Cipher Spec
5783	86.306417	192.168.100.121	192.168.100.121	TLSv1.3	177	Application Data
5785	86.313784	192.168.100.121	192.168.100.121	TLSv1.3	144	Change Cipher Spec, Application Data
5786	86.317194	192.168.100.121	192.168.100.121	TLSv1.3	146	Application Data
5787	86.317293	192.168.100.121	192.168.100.121	TLSv1.3	496	Application Data
5788	86.414896	192.168.100.121	192.168.100.121	TLSv1.3	192	Application Data
5789	86.435208	192.168.100.121	192.168.100.121	TLSv1.3	1862	Application Data, Application Data
5790	86.435426	192.168.100.121	192.168.100.121	TLSv1.3	89	Application Data

Packet Details:

- Ethernet II, Src: Intel E810 (82:55:0A:00:02:00), Dst: Intel E810 (82:55:0A:00:02:00)
- Internet Protocol Version 4, Src: 192.168.100.121, Dst: 192.168.100.121
- TCP Segment Len: 5648
- Sequence Number: 651184 (relative sequence number)
- Sequence Number (raw): 1348391047
- [Next Sequence Number: 6516832 (relative sequence number)]
- Acknowledgment Number: 7225 (relative ack number)
- Acknowledgment Number (raw): 3533327220
- 8181 = Header Length: 20 bytes (5)

Raw Data (Hex Dump):

```

0020  64 79 01 bb 04 d0 50 5e 08 87 d2 9a 4b 74 50 18 dy...P...KTP...
0030  00 af ac 51 00 00 e5 6e f8 a5 dc ca 33 67 a6 90 QLO...n.E.3g...
0040  62 28 4e 9d df 43 fd 39 dc 00 e4 69 07 bd 54 e5 (N.C.9...I.T...
0050  06 ef 1f 56 08 0c 48 83 69 93 7f 2e 40 bc b4 7 r...V..H..I...T.G...
0060  19 fe f3 aa 19 54 45 db 7f 6e f0 20 a3 b7 bf 47 t...T.K....G...
0070  25 c5 c3 cb 77 ea e0 1f ee c5 56 71 5a f4 c1 f3 S...w...WQZ...
0080  21 b8 a0 b3 c8 9e 8d 71 1a ff e4 ec 4d 67 d b ad i...>...o.C.M.m...
0090  72 dd 55 9c ab 70 b9 71 6b a9 4b 73 40 e7 13 eb r.U...y.kKSb...
00a0  4f 34 16 3f 9a 16 fe 0b c8 95 34 56 23 9d 8e 6c O4?.....Z.W-1...
00b0  ad bc 26 e4 3e 1b a9 b8 5a d0 97 39 f4 bc 80 98 &...>...Z.9....
00c0  00 b4 c4 6d 33 ff 8f bf 8d 9a 20 8a bf f4 18 ch m3.....t...
00d0  f7 66 1d 28 49 91 8f cd 5d ff 5d 78 c5 7c 2d 3f F[...I.]x{[k}...
00e0  fd eb 14 81 e0 69 84 6b dc 6d 82 46 c5 a9 05 50 -i.k.m.F.P...
00f0  fd 68 ae bd b8 75 a0 3c 55 96 48 ac 8c 76 cf h-u.c-U.H.-t...
    
```

Answers to the questions

1. Approximately 1500 packets were captured during the experiment.
2. Let's list three protocols observed during the experiment:
DNS, HTTP, TLS
3. Encryption is important when browsing the internet because it protects sensitive information transmitted over the Internet. Without encryption, data such as passwords, personal information, or financial data could be intercepted and read by attackers.
4. Let's show the types of information could attackers obtain from unencrypted traffic:

An attacker could access several types of sensitive information, including login credentials, personal data, session cookies, email addresses, requests sent to websites, or even the pages visited by the user.

Lab Report

This experiment clearly shows that network traffic can be easily captured and analyzed with accessible tools. It also demonstrates that weak passwords represent a major vulnerability and that unencrypted communications can expose sensitive information. Therefore, the use of encryption and strong passwords is essential to protect users' security and privacy on the Internet.